



1. Project Data

Project ID

P150395

Project Name

TN Sustainable Urban Development Project

Country

India

Practice Area(Lead)

Urban, Resilience and Land

L/C/TF Number(s)

IBRD-84880

Closing Date (Original)

31-Mar-2022

Total Project Cost (USD)

352,729,993.47

Bank Approval Date

31-Mar-2015

Closing Date (Actual)

31-Mar-2023

IBRD/IDA (USD)
Grants (USD)

Original Commitment

400,000,000.00

0.00

Revised Commitment

352,729,993.47

0.00

Actual

352,729,993.47

0.00

Prepared by

Cynthia Nunez-Ollero

Reviewed by

Vibecke Dixon

ICR Review Coordinator

Avjeet Singh

Group

IEGSD (Unit 4)

2. Project Objectives and Components

a. Objectives

According to the Financing Agreement (FA, p 5) and the Project Appraisal Document (PAD, paragraph 11), the Project Development Objective (PDO) of this Tamil Nadu Sustainable Urban Development Project was "to improve urban services delivery in participating Urban Local Bodies (ULBs) in a financially sustainable manner and to pilot improved urban management practices in selected cities."

This review will parse this objective into two:



- To improve urban services delivery in participating Urban Local Bodies (ULBs) in a financially sustainable manner
- To pilot improved urban management practices in selected cities.

b. Were the project objectives/key associated outcome targets revised during implementation?
No

c. Will a split evaluation be undertaken?
No

d. Components

1. Results Based Grants for Urban Governance: (US\$61 million, of which IBRD US\$55 million at appraisal; IBRD financing reduced to US\$54.5 million at restructuring; US\$57.41 million, of which IBRD US\$51.67 million, actual) This component was to finance results-based grants to three 'Model Cities' - Erode and Vellore municipal corporations, and Hosur municipality. These grants were to be distributed on a pilot basis to finance improvements in four urban management areas: (i) empowerment and organizational capacity; (ii) spatial development planning; (iii) sustainable finances; and (iv) e-governance and public disclosure. In addition, this component was to finance project management support and consultant services for the Directorate of Municipal Administration (DMA) to implement this component. At the 2020 restructuring, US\$0.50 million allocated for project management and technical assistance (TA) to DMA was reallocated to increase the TA for Urban Local Bodies (ULBs) under Component 2 below.

2. Investments in Urban Services (US\$504 million, of which IBRD US\$323 million at appraisal; IBRD financing increased to US\$516 million at restructuring; US\$462.81 million, of which IBRD US\$290.51 million, actual) The Tamil Nadu Urban Infrastructure Financial Services Limited (TNUIFSL) was to implement this component. The TNUIFSL is a public limited company of the Government of Tamil Nadu that served as a Public-Private Partnership with equity participation from banks and financial institutions including the Industrial Credit and Investment Corporation of India (ICICI) Bank, the Housing Development Finance Corporation (HDFC) Limited, and Infrastructure Leasing and Financial Services (IL &FS) Limited. This component was to finance: (i) investments in urban services, including water, sewerage, municipal solid waste, urban transportation, sewage management, and storm water drainage (SWD), and support project management and supervision; (ii) a Credit Rating Enhancement Fund (CREF) to pilot instruments such as municipal bonds and other market-based financing instruments that ULBs in Tamil Nadu may use to mobilize resources beyond budgetary sources; and (iii) technical assistance (TA) to ULBs, the Greater Chennai Corporation (GCC), and the TNUIFSL. The TA for ULBs was to help prepare and implement subprojects, including environmental and social (E&S) mitigation actions, public-private partnership (PPP) arrangements, and implement credit enhancement measures. The TA to the GCC was for urban flood risk mitigation. The TA for TNUIFSL was for institutional development. Four changes occurred during implementation. First, in July 2015, after the World Bank approved this loan (see Dates below), the Securities and Exchange Board of India (SEBI) adopted the 2015 Issue and Listing of Debt Securities by Municipalities Regulation. This law stated that only a municipality or a corporate municipal entity may issue debt securities or bonds, but not trusts such as the Water and Sanitation Pooled Fund or TNUDF, as designed under the CREF. TNUIFSL was unsuccessful in requesting an amendment. The tight domestic credit markets at that time also meant that commercial banks loans were unviable. At the project midterm review (MTR) and in the 2020 restructuring, CREF was dropped (see Section 4 Efficacy below). Second, the US\$20 million allocated to CREF was redistributed to urban investments and TA under this



same component. Third, the government increased its contributions by US\$9 million and increased the overall allocation for this component from US\$504 million to US\$516 million (US\$9.0 million from the government and US\$0.5 million from component 1 above and US\$3.5 million from component 3 below). Fourth, in the 2021 restructuring, a new COVID-19 related response subcomponent was added taking US\$5.0 million from urban investments to finance oxygen supplies, medical and testing equipment/kits and supplies, essential equipment for case management, essential pharmaceuticals, and training and capacity building related to COVID19 care.

3. Urban Sector Technical Assistance (TA) (US\$34.0 million, at appraisal of which IBRD was US\$21.0 million; US\$13.88 million, actual of which IBRD US\$9.55 million) This component was to be implemented by DMA and was to finance TA for ULBs: (i) to develop next-generation municipal e-governance and geographic information systems (GISs); (ii) training, sector studies, operations and maintenance (O&M), and strengthening of public financial management for institutional development and capacity building. This component was also to finance project management and DMA's incremental operating costs. In 2020, the project realized savings of US\$3.5 million from lower than-anticipated training expenditures and incremental operating costs. The amount was reallocated to Component 2 above.

e. Comments on Project Cost, Financing, Borrower Contribution, and Dates

Project Cost: The total project cost was US\$400.00 million. The loan disbursed US\$352.73 million. The balance of US\$47.27 million was cancelled (ICR, paragraph 56).

Financing: The International Bank for Reconstruction and Development (IBRD) financed this loan.

Borrower Contribution: The government committed US\$200.00 million in contributions, added US\$9.0 million at restructuring, and disbursed US\$182.37 million at closing. The balance of US\$26.63 million was undisbursed because of the reduced number of project-affected families to be resettled requiring assistance.

Dates: The loan was approved on March 31, 2015 and made effective on August 21, 2015. The Mid Term Review (MTR) was conducted on May 6, 2019. The loan was to close originally on March 31, 2022 but was extended (see below) by an additional 12 months to close on March 31, 2023. The project was restructured three times:

- On June 30, 2020 to reallocate loan savings of US\$2 million from Components 1 (results-based grants) and 3 (urban sector TA) for priority urban sector TA (component 3), cancel US\$18 million for CREF (Sub-Component 2) to finance cost over-runs under component 2 and make the corresponding changes to the targets in the indicators of the results framework.
 - At the PDO outcome levels:
 - indicator 1: Increased the target number of direct project beneficiaries from 2.0 million to 2.2 million, of which the target of 40 percent female was unchanged.
 - indicator 2: Added the target of 95 percent repayment rate on sub-loans by TNUDF to ULBs.
 - indicator 3: Clarified the target of the cumulative Increase in own source revenues (OSR) over the previous year of 'Model Cities' expressed in INR crores. The target cumulative increase in Erode and Vellore of Rs. 31.93 crore each and in Hosur an increase of Rs.15.96 crore over and above the OSR of 2016-17.



- At the output levels:
 - reduced the end target of “participating ULBs with updated GIS maps” from 90 to 30 ULBs.
 - increased the end target of “participating ULBs with operational new e-governance systems” from 90 to 135.
 - revised the end target for “Improved ULB empowerment in the ‘Model Cities’” from “Each ULB has adopted a sustainability plan for the technical expert cell’s functions and responsibilities after the Project’s closing date” to “Each ULB has appointed a high-quality Technical Expert Cell.”
 - revised the end target for “Improved ULB citizen engagement in the ‘Model Cities’” from “Each ULB has established an online platform to exchange information with citizens regarding key service delivery parameters” to “Each ULB has implemented all the e-governance modules as identified by the Chennai Metropolitan Area Administration (CMA).”
- On June 9, 2021 to add a new disbursement category for the COVID-19 expenditures and reallocate costs among components, with no changes to the results framework. Costs were reallocated because (a) funding for urban investment subprojects was increased to cover the increased costs of existing subprojects and to add new subprojects to utilize loan savings; (b) the cancellation of the CREF subcomponent due to a change in regulation; 2 (c) the addition of the COVID-19 pandemic emergency response subcomponent; (d) lowering the target related to GIS mapping given the capacity constraint in DMA and ULBs, and increasing the target related to e-governance system to scale up impacts; and (e) the increase in project duration to complete construction activities that were delayed mainly due to the COVID-19 pandemic, the 2021 state elections, and flooding in Chennai in November 2021. The cancellation of the CREF activity does not affect the PDO outcome in a major way, as the CREF activity represented only 3.3 percent of the overall project cost, and there were other activities under Component 1 and 3 to support increase in ULBs’ revenue collection and hence enhance their financial sustainability. Overall, these changes expanded the scope of relevant activities (particularly urban investments, COVID-19 response, and e-governance systems), while removing an activity that could not be implemented because of national regulatory changes.
- On February 8, 2022 to extend the loan closing date to complete ongoing activities.

Split Rating: A split rating does not apply because the PDOs remained unchanged, and the project scope was scaled up. While the CREF activity and its associated indicator were dropped, other original indicators - related to loan repayment and OSRs - sufficiently captured the outcome of ‘improve financial sustainability.’ In addition, reallocated resources expanded the project’s scope - increased the number of target beneficiaries, increased OSR generation, and expanded e-governance systems. According to the World Bank Guidance: ICR for Investment Project Financing (IPF) issued on December 9, 2021, para. 60, regarding projects with expanded scope: “If the project became overall more ambitious, a split rating is not applied [...]. The operation can be assessed based on the more ambitious revised outcomes and outcome targets.”

3. Relevance of Objectives

Rationale



Context: Tamil Nadu is India's most urbanized state. Its urban population has increased from 24.4 percent in 1951 to 48.5 percent in 2011. 12 corporations, 124 municipalities, and 528 town panchayats comprise the state of Tamil Nadu. *Panchayat* means "assembly" (*ayat*) of five (*panch*). Traditionally, panchayats consisted of respected elders selected by the local community to settle disputes both between individuals and villages. Chennai, the capital of Tamil Nadu is the sixth most populous city in India with an estimated population of 4.9 million, in an area of 426 square kilometers after a 2011 expansion. The Corporation of Chennai merged peri-urban areas consisting of 9 Municipalities, 8 Town Panchayats and 25 Village Panchayats and has an estimated population of 9 million. Service gaps were noted for the urbanizing areas of Tamil Nadu. Sewerage and storm water drains covered less than 30 percent of the city. Less than 10 percent of solid waste was treated and disposed. City governments needed to generate financial resources, create jobs, and improve urban governance. The Government of Tamil Nadu noted in its Urban Vision 2023 that urban local bodies (ULBs) required an estimated US\$42 billion in investments during 2012–2023 to meet its urban challenges.

Country Plans: The Government of India launched several urban development programs such as the '100 Smart Cities', 'Atal Mission for Rejuvenation and Urban Transformation' (AMRUT), and 'Swachh Bharat (Clean India) Mission' (SBM). The PDO remained relevant to Vision Tamil Nadu 2023 "to promote and facilitate the development of inclusive and sustainable cities." This project focused on water supply service improvement, reuse of wastewater, e-governance, and strengthening of the financial and administrative capacity of ULBs also closely aligned with the objectives of India's flagship missions AMRUT 2.0 and SBM 2.0 (2021–2026), as well as the Tamil Nadu Urban Vision 2031 that aims to achieve universal access to basic infrastructure and services in all ULBs, improve environmental sustainability, and strengthen ULBs' institutional capacity.

World Bank Country Partnership Framework: At closing, the PDO remained relevant to the WBG India Country Partnership Framework (CPF) for FY18–22 (ICR, paragraph 21). The PDO contributed to Area I: Promoting resource-efficient growth. The 2018 Systematic Country Diagnostic noted that efficient cities contributed to resource-efficient growth for India. The PDO supported making ULBs more responsive to its constituents by investing in basic services and improving urban governance. The PDO was aligned with the World Bank's goals of eliminating poverty and promoting shared prosperity by delivering water supply services to underserved populations, including low-income people in the cities and peri-urban areas of Tamil Nadu.

World Bank Experience in the Country and in the Sector: The World Bank has a long engagement in Tamil Nadu and is the repository of international good practices in urban management. This project was to demonstrate innovation in three areas: (i) addressing the gaps in urban services, while focusing on improving environmental sustainability; (ii) setting up a credit enhancement facility to increase ULBs' access to the capital market, based on TNUDF's experience in issuing municipal bonds and similar interventions in the United States and elsewhere; and (iii) piloting a results-based financing approach with disbursements as incentives to improve urban governance in selected ULBs. Lessons from other World Bank projects, such as the Pakistan Punjab Cities Governance Improvement Project, the Chennai City Partnership: Sustainable Urban Services Program (CCP, P175221) and preparation of the Tamil Nadu Climate Resilient Urban Development Program (TNCRUDP, P179189).

The PDO was formulated at an outcome level with the financial sustainability dimension. The Theory of Change (ToC) in the ICR indicated that the improvement of urban service delivery and management practices was expected to lead to improved living conditions for urban residents, improved environmental



sustainability and resource efficiency, and improved transparency and citizen engagement. Both the ToC and the project design showed that the project was expected to contribute to these higher-level objectives.

Overall, the relevance of the PDO is rated High. The PDOs were relevant to both the country plans for Tamil Nadu and the Urban Local Bodies (ULBs) as well as the World Bank's partnership framework for the country and the sector.

Rating

High

4. Achievement of Objectives (Efficacy)

OBJECTIVE 1

Objective

To improve urban services delivery in participating Urban Local Bodies (ULBs) in a financially sustainable manner.

The 30 ULBs include 14 municipal corporations, 15 municipalities, and one town panchayat. Two municipalities, Pallavaram and Pammal, were merged with the Tambaram Municipal Corporation.

Rationale

Theory of Change: No explicit Theory of Change (ToC) was outlined at appraisal. A retrospective TOC was prepared for the ICR. The activities were designed to deliver demand-based infrastructure investments. These infrastructure investments were to improve the urban environment, emphasized 'sustainability' (expressed as no less than 40 percent in value). ULBs were to be selected using the Tamil Nadu Urban Development Fund (TNUDF) framework criteria.

Inputs were to be the training and resources to build capacity on public financial management such as the preparation of Own Source Revenue (OSR) implementation plans. These plans were to include the financing of the operations and maintenance (O&M) needs of completed assets, delivering services to ULB residents, restoring water bodies, treating, and reusing wastewater.

Outputs were to be the physical measures of the improved urban services such as improved or newly constructed roads, installed energy-efficient streetlights, improved drainage, and new water and sewerage connections. Outputs were also to include a municipal solid waste system in place, constructed stormwater drainage, restored water bodies, and a new Tertiary Treatment Reverse Osmosis (TTRO) plant and conveyance main. ULBs were to determine these investments based on demand. Outputs to show improved financial sustainability of participating ULBs' were to include OSR implementation plans, an e-governance system - Urban Tree Information System (UTIS) to track revenue collection, and updated GIS maps for property tax assessments. Some targets were determined during implementation since ULBs were expected to prepare these investments as part of the demand-based design.



Outcomes were to be improved services delivered to gender-segregated direct project beneficiaries. Outcomes were to also include the increase in ULB resources mobilized from own source revenues (OSR) and non-budgetary sources. The outputs from the demand-based investment plans that ULBs prepared were causally linked to these outcomes.

Critical assumptions: The TOC included the following critical assumptions that, if realized, were to enhance the achievement of the PDO: (i) investments reflected the beneficiary's needs; (ii) ULBs maintained constructed assets; and (iii) infrastructure services were affordable.

The ToC is clear. The causal link between inputs, outputs, and outcomes is logical. ULB capacity is built, ULBs generate demand-based investment plans with citizen participation, priority investments are made, resources are mobilized, and improved services are delivered. The outcomes were causally attributed to the inputs and outputs of the project.

OUTPUTS:

- 30 ULBs (original target was unknown at appraisal, the target of 30 determined at implementation according to the email from the Task Team, target achieved) received result-based grants to implement reforms in empowerment and organizational capacity, spatial and development planning, e-governance, public disclosure and citizen engagement, and OSR planning.
- Completed the first TTRO wastewater treatment plant in Tamil Nadu **as targeted**. The plant treated and supplied 30 mld of used water through 28.50 km conveyance main to 11 industries in the Manali area outside Chennai city. The TTRO plant commenced operation in 2019 and started selling recycled water to industrial consumers, who reported a 17 percent decrease in manufacturing costs because of cheaper water from the TTRO.
- The project facilitated the creation of a Public Finance Management Reform Cell in the Directorate of Municipal Administration (DMA) **as targeted**. The cell provided strategic inputs and technical support to ULBs to implement public finance-related reforms in the ULBs. The cell developed financial indicators and ranking systems of ULBs and suggested improvements in the accounting records submitted by ULBs in UTIS, including sharing the reports with the public. UTIS is a web-based workflow application software that integrates 29 modules such as birth and death registration, property tax, water charges, trade licenses, building plan management, accounting system, revenue, and human resources management (ICR, footnote 11). The project also provided training on public finance management to the participating ULBs.
- Urban Sector Technical Assistance (TA): Completed capacity-building activities, including e-Governance systems implemented through the 'Urban Tree Platform' that is now fully operational in 135 municipalities in the state. 26 training and capacity building activities were conducted for 37,819 ULB staff. The critical TA for Chennai flood warning and decision support system was delayed because of delayed clearances from the Ministry of Defense. The Geographic Information System (GIS) related activity for ULBs required was to be completed during the 12 month an extension.
- 30 ULBs constructed and upgraded basic infrastructure. The Task Team clarified to IEG that the targets for these outputs were determined during implementation when the ULBs enjoined citizen participation to formulate demand-based investment plans. The Task Team provided additional information for some but not all targets by ULBs to IEG.



- Access to improved water supply: The primary source of water used to be stand posts, public fountains, and water tankers. Dependency on water tankers decreased from 18.9 percent in 2015 to 0.8 percent in 2022 (ICR, footnote 6).
 - 148,411 households were provided new water connections (baseline was 343,313 households, original target was 240,319 households or 70 percent, achieved 62 percent **not achieving** target). The proportion of new or rehabilitated water and sewer connections against aggregate targets among participating ULBs was not achieved because of implementation delays in Nagercoil, which represented 18 percent of the targeted households' water connections. This was reported to have started operation in September 2023. Delays were due to the COVID-19 pandemic, contractors unable to mobilize labor and materials, and the added time for issuing the revised administrative sanction for the cost overruns.
- Access to improved drainage and reduced flooding:
 - Built 500.4 km of new stormwater drainage systems network, including 438.8 km in Chennai (original target was 290 km, target **exceeded**),
 - Installed Real-Time Flood Forecast System (RTFFS) for Chennai, **as targeted**. A GIS data center in GCC hosts spatial data and coordinates with other departments in their use.
- The following additional outputs were reported though there were no targets.
 - Access to improved water supply: The project installed 1,068.12 km of water supply pipelines (representing an increase of 3.1 times over the baseline of 344.52 km. The project added 79.84 million liters per day (mld) of additional water treatment capacity,
 - Access to improved roads: The project constructed and upgraded 335.88 km of all-season roads and constructed storm water drains with culverts to address delayed improvements in the past 6-10 years, lack of maintenance, and damage from water and sewerage pipes that were laid out along these roads.
 - Access to improved drainage to reduce flooding: 98 kilo liters of water storage capacity was added through the rehabilitation of ponds.
 - Improved safety and visibility: The project installed 17,301 higher luminosity energy-efficient light-emitting diode (LED) streetlights that enhanced the residents' sense of security, especially for women, the elderly, and children. The installed energy-efficient streetlights generated savings of 45 percent in energy consumption and monthly savings of INR 1.5 crores (US\$0.18 million) for GCC.
 - Improved public space through ecological conservation: The project restored 508.74 acres of water bodies, including the polluted Periya lake in the Chennai Metropolitan Area.

OUTCOMES:

- The project benefited a total of 2.8 million residents. The original target was 2 million, increased at the 2020 restructuring to 2.2 million. The revised target was **exceeded**. 49 percent of the beneficiaries were female, **exceeding** the original, unrevised target of 40 percent.
- The Chennai Real-Time Flood Forecast System (RTFFS) share real-time data and issues alerts to various stakeholders based on rainfall predictions three days in advance. The Water Resource Department use the data to monitor lake operations and river flows. The Revenue Department use the data to aid in flood relief measures. The Police and the Fire Departments use the data for flood relief



and rescue. The Education Department use the data to plan school schedules during flooding incidents. The public is also provided inundation information.

- After the TTRO started selling recycled water to industries in 2019, the Greater Chennai Corporation's (GCC's) revenue increased by INR 192 crore (US\$23.20 million), representing a surplus of INR 30 crore (US\$3.63 million), thus contributing to the city's financial sustainability.
- At the end-line survey, 68 percent of the female respondents in Chennai said there was an increased sense of safety to commute after sunset owing to the installation of streetlights. The installed energy-efficient streetlights also generated savings of 45 percent in energy consumption and monthly savings of INR 1.5 crores (US\$0.18 million) for the Greater Chennai Corporation (GCC).
- All 30 ULBs serviced their debts from year-on-year revenues, achieving a 100 percent repayment rate on TNUDF sub-loans against the original target of 95 percent. This outcome showed an Increased capacity for debt servicing.
- Investments benefited the urban poor using data from a sample of ULBs. Water supply investments in four ULBs benefited 53,357 urban poor, and road investments in 11 ULBs improved the connectivity of more than 30 slums.
- The UTIS improved revenue collection (online and offline options for citizens to pay taxes and service charges, a one-stop web portal and mobile interface in place in all 158 ULBs (including 138 municipalities and 20 city corporations) in Tamil Nadu. This achievement **exceeded** the original target of 90 ULBs and the revised target of 135 ULBs. For evidence, in 2018, property tax collection efficiency increased from 36 percent in 2018–19 to 53 percent in 2021–22, leading to an increase of 63 percent, or INR 29,059 crore (US\$3.51 billion) in property tax collection.
- The ULBs' use of the Urban Tree Information System (UTIS) contributed to their financial sustainability, evident in the increase of INR 7,728 crore (US\$933.90 million) in property tax collection achieved in 23 additional ULBs (UTIS was rolled out to 158 ULBs from a target of 135). This **exceeded** the original target of mobilizing an additional INR 492 crore (US\$59.46 million). The use of the UTIS also increased the interface with citizens. This increase in transparency was evident in ULBs preparing their annual accounts within two months after the end of the financial year, submitting their accounts on time to the Fifteenth Finance Commission, and accessing grants. The is a web-based workflow application software that integrates 29 modules - such as birth and death registration, property tax, water charges, trade licenses, building plan management, accounting system, revenue management, and human resources management - in all 158 ULBs (including 138 municipalities and 20 city corporations) in Tamil Nadu. This achievement **exceeded** the original target of 90 ULBs and the revised target of 135 ULBs.
- A total of 622,328 people were provided access to wider and better-quality roads. Target was 647,532, according to the Task Team. Target was **almost achieved**. Respondents to the end line survey stated that on average, they saved about 30 minutes in travel time due to reduced congestion, and experienced improved walking and cycling experience to work, schools, and markets.
- The project provided safe and reliable piped water supply to 774,337 people in 11 ULBs (target was 1,231,412, target **not achieved**). The coverage of tap water supply increased from 49 percent before the project to 82 percent in 2022. Per capita supply achieved the state target of 135 liters per capita per day (lpcd) in the ULBs receiving investment, i.e., more than twice the baseline supply of 50 lpcd in 2011. About 91 percent of the respondents to the end line survey stated that they were satisfied with the quality of water supplied post project, against 35 percent in 2015.
- The project reduced the impact of flooding, benefiting 1.16 million people in the three ULBs where stormwater drainage investments were made. The target of 1.16 million was achieved. The project also reported on the following, though there were no targets set. In the coastal city of Cuddalore, the



storm water drains covered a wider area from 36 to 90 percent, and inundation reduced from 204 to 12 hours. In Cuddalore, all waters drain from the Gedyam River to the sea. Inflows from all secondary drains reduce the carrying capacity of the channel, leading to high water levels in the river and long-standing time for flood waters to recede. In Chennai and Thoothukudi, flooding hot spots were reduced by 50–70 percent, and the hours of inundation were reduced from 168 hours to less than half an hour.

- The project Increased property tax via GIS mapping. This resulted in an additional revenue of INR 65 crore (US\$7.85 million) per year from 14 percent of bills verified. With all the bills verified, GCC's annual tax revenue was expected to increase by about of INR 500 crore (US\$60.42 million). A drone survey was completed in 14 ULBs. A mobile application for field survey was developed. Digitalization of GIS maps and door-to-door survey in these ULBs was not completed because DMA experienced delay in appointing a GIS specialist. The indicator 'Number of participating ULBs with updated GIS maps' did not achieve the original target of 90 or the revised target of 30. A follow-on operation, TNCRUDP, was expected to complete these activities. GIS mapping in the target ULBs was not expected to have an immediate impact on property tax collection, because revenue staff can only conduct field verification of bill variations after the GIS maps are updated. In GCC experience, this update may take over a year.
- According to beneficiary surveys and secondary data from real estate portals, infrastructure improvements under the project increased property values. This was expected to increase property tax collection by the ULBs. On average, rentals in the project areas have increased 12 percent in the last five years (the control group reported a 3 percent increase).

The ToC provided the causal attribution of the inputs and outputs that contributed to the outcomes to achieve the PDO (improved urban services delivery in a financially sustainable manner). The evidence from the end-line surveys showed beneficiary satisfaction with the improved services delivered. Evidence from the increase in resource mobilization showed the sustainable way these services were to be supported in the long term.

Overall, the efficacy of the project to achieve this objective is rated Substantial with moderate shortcomings. The project exceeded the two outcome indicators - the number of beneficiaries and the increased capacity for debt servicing. The project achieved six of the eight output targets - solid waste collection coverage, citizen engagement, and empowerment, stormwater drainage investments in Chennai, the proportion of ULBs reporting improved service delivery, and e-governance. The output targets related to GIS mapping and new or rehabilitated water and sewer connections were not achieved because of implementation delays related to cost overruns and COVID-19 disruptions.

Rating

Substantial

OBJECTIVE 2

Objective

To pilot improved urban management practices in selected cities (i.e., three Model Cities) Erode and Vellore Corporations and Hosur Municipality

Rationale



Theory of Change: A ToC was not prepared at appraisal but was reactively constructed for the ICR. This objective was designed to provide results-based grants to three pilot cities - Erode and Vellore municipal corporations and Hosur municipality - to fund investments and urban management practices. These ULBs were representative of medium-size cities with populations between 200,000 and 500,000 and were either newly upgraded or expected to be upgraded from municipalities at the time of preparation and was an opportunity to provide capacity support. Every year, grants were to be released to these cities after they met the four targets of disbursement linked indicators (DLIs) or Performance Based Conditions (PBCs). These DLIs served as incentives for the cities to improve in four urban management areas: (i) empowerment and organization capacity; (ii) spatial and development planning; (iii) sustainable finances; and (iv) e-governance, public disclosure, and citizen engagement.

Inputs were to include the training and technical assistance to assist the model cities to develop e-governance systems, upgrade GIS maps for property tax assessments and build capacity to undertake public private partnerships and improve public financial management. Inputs also included the grants that were distributed to the three model cities after they met the targets of DLIs.

Outputs were to be an installed e-system to monitor revenue generation, called Urban Tree Information System, upgraded GIS maps to improve property tax assessments, and OSR improvements plans for improved resource mobilization.

Outcomes were to be improvements in the financial management of the Model Cities based on the effective use of the UTIS. The UTIS replaced the manual systems. The Model Cities would provide real-time information on ownership and conditions of assets to tap into market resources. Other outcomes included improving solid waste management services, increases in property tax receipts due to improved assessments assisted by the upgraded GIS mapping; and increases in revenues from non-budgetary sources, such as fees and permits.

Critical assumptions: The same critical assumptions noted in the Theory of Change under objective 1 applied to this TOC.

The ToC was clear. The outcomes were attributable to the causally linked inputs and outputs. A minor gap was a lack of indicator associated with the resulting leverage that would have tapped into private sector participation in delivering services because of the improved financial management of the Model Cities.

OUTPUTS:

- The three model cities received grants of INR 357 crores (US\$43.14 million) and achieved a disbursement rate of 94 percent of the original target of US\$45.9 million, **almost achieving** the target of 100 percent. The Task Team clarified to IEG that the achieved disbursement rate refers to the amount of fund disbursed against the revised allocation for the Results Based Grants for Urban Governance. The grants were invested in improving service delivery and develop revenue-generating amenities such as commercial complexes, community halls, parks, and gyms.
- The Model Cities developed revenue improvement plans and **achieved** OSR targets for the first four years. Before 2020, the ULBs' revenue staff conducted field verifications and reported the expansion of the property tax base, increased property tax collections and non-tax items such as rent from markets, community halls, plot regulation charges, and solid waste user charges. Beginning in 2020,



the annual targets were **not achieved** because the state government rolled back the property tax rate increase in place and lowered tax filing and collection during the COVID-19 pandemic.

OUTCOMES:

- The 'Model Cities' achieved an increase of INR109 crores in OSR over the previous year after implementing its respective OSR improvement plans (US\$13.17 million) in 2021–22 over the OSR of 2016–17 (**exceeding** the target of INR79.82 crore). The baseline was INR72 crores in FY13. The original target was an increase of INR22.5 crores. The original target was **exceeded**. The target was revised at restructuring to specify targets for each Model City (i) Erode and Vellore Corporations were to increase OSRs over the previous year by Rs31.93 crore each in the end of project over and above the OSR of 2016-17; (ii) Hosur Municipality was to show cumulative increase of OSR amounting to Rs15.96 crore in the end of Project over and above the OSR of 2016-17
- The three Model Cities **achieved** the targets for the three DLIs related to Improved technical and managerial capacity for urban management - empowerment and organizational capacity (DLI 1), spatial and development planning (DLI 2), and e-governance, public disclosure, and citizen engagement (DLI 4). The cities were the first to implement UTIS. The Model Cities established Technical Expert Cells and developed Multiyear Capital Investment and Urban Design Improvement Plans. The cities invested about US\$24 million in grants and OSR for 11 sub-projects. The Model Cities reduced staff vacancies to less than 5 percent of sanctioned posts (**exceeding** the target of 20 percent); resolved about 80 percent of grievances within the time limit set in their Citizen's Charters (target 80 percent); and achieved 90 percent of the Ministry of Urban Development Service Level Benchmarks Index related to solid waste household collection coverage, **achieving** the target.
- There was a 90 percent average increase on Service Level Benchmarks (SLB) Index for the 'Model Cities' related to solid waste household collection coverage (baseline was 65 percent, target 90 percent, target **achieved**).
- Repayment rates on sub loans by TNUDF to ULBs (baseline 100 percent, target 95 percent, **achieved** 100 percent, exceeding the target). This was initially an Intermediate Result Indicator of PAD and was changed to a PDO indicator in 2020. The e-governance system implemented contributed to an increase in revenue collection, GIS mapping improved the property tax base, and strengthened the financial management capacities of the ULBs.

The objective of improved urban management practices in the Model Cities was achieved from the causally linked inputs and outputs outlined in the ToC. The use of the UTIS boosted the capacity of the Model Cities to deliver urban services (solid waste management), increase own source revenues (property tax receipts) and non-budgetary revenues (fees and permits) and increase transparency and inclusiveness in planning and management. Overall, the efficacy of the project to achieve this objective is rated Substantial. Systems were in place to foster improvements in urban management practices in the three model cities.

Rating
Substantial



OVERALL EFFICACY

Rationale

The first objective to 'improve urban services delivery in participating ULBs in a financially sustainable manner' was substantially achieved with moderate shortcomings especially on the water supply and sewer connections which were not achieved due to implementation delays. The second objective of improved urban management practices in the Model Cities was substantially achieved. Overall, the efficacy of the project to achieve both objectives is rated Substantial.

Overall Efficacy Rating

Substantial

5. Efficiency

Economic Efficiency: At appraisal, the project adopted a framework approach for project investments under the second component, Investments in Urban Services. Prior to implementation, municipal subprojects under this component were subjected to economic analysis. The stormwater drainage (SWD) subproject in the Greater Chennai Corporation (GCC) was also subject to cost-benefit analysis. The expected benefits were tangible (direct and indirect) socioeconomic benefits from avoided flood damage; and intangible benefits from improved environment and living conditions. These intangible benefits were not quantified. The economic rate of return (ERR) was estimated at 15.5 percent using a 12 percent discount rate. Sensitivity analysis led to a range in ERRs from 13.48 to 15.3 percent under adverse scenarios.

In closing, the cost-benefit analysis methodology used at appraisal was replicated in the investment subprojects - the stormwater drainage (SWD), water supply, road improvements, street lighting, and tertiary treatment reverse osmosis (TTRO). Estimated economic benefits and actual costs at closing showed EIRRs between 15 and 21 percent using the same 12 percent discount rate used at appraisal. The EIRR for SWD at completion was 16.40 percent, compared to the 15.5 percent at appraisal (ICR, paragraph 43).

Financial Efficiency: At appraisal, the financial appraisal of the subprojects was divided into two: (i) at the subproject level, water and wastewater subprojects were assumed **not** financially viable based on subproject revenues. At the ULB level, all 30 ULBs were assessed in the aggregate to assess whether subprojects were financially sustainable, and this was measured by borrowing capacity. The overall borrowing capacity of all ULBs over five years was estimated to be over US\$600 million. The financial analysis concluded that this meant ULBs were able to absorb about US\$150 million in sub-loans. In closing, the same method indicated that in ULBs that implemented water supply subprojects (using data from 21 of 30 participating ULBs), the tariff collected increased from 32 to 81 percent. This rate indicated that the capacity of these ULBs to meet water supply operations and maintenance (O&M) costs to be from three sources: (i) its tariff collections, (ii) the share of the property tax earmarked for water supply, and (iii) from ULB general funds. At the ULB level, the analysis of 15 of 21 ULBs showed a ratio of less than 1 for total expenditure to total revenue. This meant strong revenue flows. All 115 ULBs showed a debt service ratio (DSR) of less than 15 percent, showing that debt service could be met by annual revenues. The World Bank determined an affordability threshold of 5 percent of the average household income. The average monthly flat fee in the ULBs was less than INR 300 (US\$3.63) for water or about 1–2 percent of the average household income (ICR, paragraph 44).



Administrative and Operational Efficiency: Implementation delays experienced in 2020–2021 resulted in a 12-month extension of the project. These delays came from (i) COVID-19 pandemic restrictions resulted in the stoppage of works and diversion of ULB staff working on the project, leading to an overall delay of about 9–10 months; (ii) the 2021 elections in the state, the resulting new government, flooding in Chennai in November 2021; (iii) internal government approvals taking much longer than anticipated, while clearances were also required from multiple line departments for the water supply lines to pass through the railways; and (iv) changes in design that resulted in cost escalations and required further approvals. Some water supply subprojects had to change the siting of water storage structures because of protests from the public, inappropriate site characteristics (such as soil strata), or the presence of railway lines, resulting in delays and cost escalations. Loan savings were used to complete additional works (roads in ULBs and links in the Chennai stormwater drains) during the extension. The government disbursed 90 percent of counterpart funds.

Overall, with a reasonable economic rate of return for the investments made under component 2, and the added activities financed by loan savings completed during the extension, efficiency is rated substantial.

Efficiency Rating

Substantial

a. If available, enter the Economic Rate of Return (ERR) and/or Financial Rate of Return (FRR) at appraisal and the re-estimated value at evaluation:

	Rate Available?	Point value (%)	*Coverage/Scope (%)
Appraisal	✓	15.50	30.00 ☐ Not Applicable
ICR Estimate	✓	16.40	30.00 ☐ Not Applicable

* Refers to percent of total project cost for which ERR/FRR was calculated.

6. Outcome

The relevance of objective is rated Substantial. The efficacy of the project to achieve objective 1 is rated Substantial with moderate shortcomings because some significant output targets were not achieved. The efficacy of the project to achieve the second objective is rated Substantial. Efficiency is rated Substantial because of reasonable rates of return at closing. The overall outcome is rated Satisfactory.

a. Outcome Rating

Satisfactory

7. Risk to Development Outcome



The following pose risks to development outcome:

- **Risk from inadequate financial resources for Operations and Maintenance (O&M) of completed infrastructure assets.** In this project, ULBs raised water tariffs to cover the O&M and debt servicing costs of their water supply systems. Public resistance delayed implementing tariff increases. Collection efficiency was suboptimal. To mitigate this risk in the future, ULBs initiated measures such as field staff conducting door-to-door campaigns to raise public awareness, increase effort at water tariff collections, and collect penalties for late payment. ULBs may consider signing service contracts with third parties for O&M of water supply systems with payments linked to meeting service standards. ULBs may develop an asset management system with an O&M implementation plan for non-revenue generating infrastructure assets.
- **Risk from natural hazards and climate change.** GIS mapping and the Real Time Flood Forecasting System (RTFFS) were introduced to assist ULBs to plan, reduce impact on completed infrastructure assets, maintain the reliability of urban services, and respond to disasters resulting from hazards brought by flooding, drought, and other environmental risks. Continued use of these systems, with commitment to periodic updates, would mitigate this risk.

8. Assessment of Bank Performance

a. Quality-at-Entry

The project and the PDOs were aligned with the government's national and state priorities. The state government committed a considerable amount of contributions. The participating ULBs demonstrated their commitment by identifying available government land for infrastructure. The equivalent of 30 percent of subproject costs were ready for procurement by the time of Bank approval as required by the national government. Implementation arrangements were adequate. The project used agencies that participated in earlier World Bank-financed projects and were familiar with World Bank policies and procedures. The Bank team designed the project informed by lessons from prior World Bank engagement in the sector in Tamil Nadu. These included (i) a state-level urban development agency may reach more ULBs using a 'Municipal Fund' to facilitate reforms and innovations; (ii) having bid documents for subprojects ready for procurement equivalent to 30 percent of project cost by the time of Board approval strengthens a readiness to implement; and (iii) sites ready and available for civil works once contracts were awarded reduces time and cost overruns. Experience in earlier TNUDF-issued municipal bonds, and the US municipal bonds / bond banks experience informed the credit enhancement sub-component design. The DLI mechanism of the urban governance component benefited from other Bank project experience, e.g., the Pakistan Punjab Cities Governance Improvement Project (PAD, paragraph 19).

The World Bank task team included operational and technical specialists with experience in urban development projects in India. The task team conducted comprehensive institutional assessments of the ULBs on urban services, governance, and finance, which informed the technical designs. To improve ULBs' capacity in O&M, Component 1 provided incentives to the Model Cities to prepare Multiyear Capital Investment Plans that would include an assessment of the O&M needs of various urban services and the identification and mobilization of funding sources. During subproject appraisal, TNUIFSL also supported ULBs in assessing the financial sustainability of water supply services and in setting the appropriate level of water tariff to meet O&M needs, and the recommended actions such as phased



increase of tariffs and expected improvement in collection efficiencies were included as part of TNUISFL's loan/grant conditions for ULBs. The PDO statement was realistic, and the project's interventions supported its achievement. The RF was overall well designed.

However, there were shortcomings in the risk assessment on procurement capacity, and technical capacity in GIS related activities. The broader regulatory environment on municipal finances could have been more closely examined prior to the inclusion of the credit enhancement subcomponent in the project. Overall, the assessment of Bank performance at entry is rated Satisfactory.

Quality-at-Entry Rating

Satisfactory

b. Quality of supervision

The World Bank team conducted 12 implementation support missions over the eight-year implementation period with technical missions in between. The task team leader and most team members were based in New Delhi and regularly interacted with country counterparts. Fiduciary, environmental, and social safeguards specialists participated in the missions and conducted additional just-in-time technical missions to address weaknesses. The World Bank team consisted of appropriate technical staff to support the project. Additional experts were mobilized as needed, including during the MTR in 2019. The Bank team used the restructurings to adjust targets and indicators brought by policy changes (e.g., the CERF sub-component), cost overruns, delayed procurement, or add activities to use US\$32 million in loan savings to ensure that the objectives remained achievable. The added activities were considered because these did not require land acquisition, resettlement or carried environmental or social safeguards risk. Savings came from cancelling the CERF subcomponent, the Hosur sewerage subproject, and exchange rate gains. The Bank team helped the government to respond to the fiscal crisis due to the COVID-19 pandemic.

Overall, the Bank performance at supervision is rated Satisfactory.

Quality of Supervision Rating

Satisfactory

Overall Bank Performance Rating

Satisfactory

9. M&E Design, Implementation, & Utilization



a. M&E Design

The Theory of Change showed clear causal links between project activities, outputs, and the expected outcomes. The PDOs were clearly specified and parsed in two. The intermediate results indicators were adequate to capture the contribution of the operation's components and outputs to achieve PDO-level outcomes. These indicators were specific, measurable, achievable, relevant, and time bound. Baselines and targets were available for all indicators although one outcome indicator was expressed as a value rather than a rate of increase. The M&E design and implementation arrangements were institutionally embedded in the various implementing entities. TNUIFSL was to develop a management information system (MIS) for monitoring physical and financial progress, and ESMF subproject compliance (Component 2, ICR, paragraph 60).

b. M&E Implementation

TNUIFSL implemented the M&E system as designed. Each ULB, the DMA, and TNUIFSL had dedicated staff in charge of M&E to monitor and collect data on indicators. TNUIFSL/TNUDF reported on indicators in the results framework to assess the physical, procurement, and financial progress in each ULB and identify outstanding issues using the two MIS' systems to integrate investment planning, project progress monitoring, contract management, withdrawal applications and payments, capital budgeting, and financial management. An independent verification agency (IVA) verified results on the DLIs of the Model Cities. TNUIFSL compiled and DMA verified information from ULBs and the IVA. TNUIFSL submitted quarterly reports to the World Bank. TNUIFSL, the DMA, and their regional offices conducted joint monthly on-site supervision regularly to the participating ULBs.

The 2020 restructuring revised two outcome indicators - one increased its target, the other specified targets per Model City - and a new one added, moved up from output to outcome indicator. One sub-component was no longer implementable and was cancelled because of a change in government policy. The related output indicator was dropped. Other target values of output indicators were adjusted, and COVID-19 related output indicators were added at the 2021 restructuring.

The beneficiary end-line survey was completed in March 2023. Beneficiaries were consulted during project design and implementation. Stakeholders helped identify sites for overhead and underground water supply tanks in ULBs where government lands were not available. Public consultations identified road stretches and inundated streets in need of improvement. Model Cities beneficiaries helped identify areas for urban design improvements financed by the grants received. Residents and shopkeepers along the Chennai storm water drain canals moved to new resettlement site. GCC reduced the width from the earlier 13 to 10 meters, to avoid existing structures and reduced the resettlement impacts from the originally anticipated 330 to 91 families (ICR, paragraph 58). However, shortcomings in indicators were not addressed. For example, indicators that would have informed regulatory readiness of the sub-component CERF (see Section 8, Bank Performance at Entry above).

c. M&E Utilization

M&E findings were communicated to various stakeholders, including TNUIFSL, the DMA, and the World Bank. M&E reports tracked progress, identified risks, delays in site clearance, and contract management. M&E data informed the midterm review and the three restructurings. M&E data provided evidence of the outcomes. M&E data served to inform activities that were not completed during this



project period to be implemented in the follow-on project, Tamil Nadu Climate Resilient Urban Development Program (TNCRUDP, ICR, paragraph 35).

Overall, the project M&E is rated Substantial. The M&E system as designed and implemented was sufficient to assess the achievement of the objectives. The links in the results chain were adequate, with moderate weaknesses in a few areas, e.g., outcome indicators (rate, as opposed to value) and targets of intermediate results indicators that could have been adjusted during restructuring, e.g., the target for GIS mapping could have been recalibrated during the restructurings.

M&E Quality Rating

Substantial

10. Other Issues

a. Safeguards

Environmental Safeguards: The project was classified as Environmental Assessment Category A (Full Assessment) (PAD, ICR, paragraph 64) and triggered two safeguard policies: Environmental Assessment (OP/BP 4.01) and Involuntary Resettlement (OP/BP 4.12). An Environmental and Social Management Framework (ESMF) was approved and disclosed in-country and in the World Bank InfoShop on March 2015. In 2020, for the COVID-19 pandemic related restructuring, the ESMF was updated and redisclosed both in-country and on the World Bank's external site in September 2022. According to the ICR, the project complied with environmental safeguards (ICR, paragraph 65). Safeguard documents, including Environmental and Social Impact Assessments (ESIAs) and Environmental and Social Management Plans (ESMPs), were implemented as designed. Quarterly Progress Reports submitted to the World Bank included updates on environmental safeguards management. Independent annual E&S audits identified gaps and corresponding corrective measures in safeguards management. The project adopted COVID-19 pandemic protocols for works and activities under execution. These included sensitizing and training of project staff, contractors, and workers on appropriate measures in relation to the COVID-19 pandemic.

Social Safeguard: The project triggered Involuntary Resettlement (OP/BP 4.12). The project complied with the social safeguard (ICR, paragraph 66). The Chennai storm water drain subproject involved involuntary resettlement of 91 project-affected families (PAFs) in 2017. The project followed its Resettlement Policy Framework (RPF). The World Bank approved the comprehensive Resettlement Action Plan (RAP). The Tamil Nadu Housing and Urban Development Board (TNUHDB) developed two resettlement sites. The GCC resettled the 91 PAFs in those sites. The GCC managed the resettlement process with regular consultations with the PAFs and integrated them with the host communities. However, government policy called for title deeds of PAF tenements to be provided after five years of resettlement. With the help of the government, GCC, TNUHDB, TNUIFSL, and the World Bank, nine of the ten PAFs resettled in the Perumbakkam resettlement site received their title deeds by September 2022. In the Gudapakkam resettlement site, 76 of the 81 PAFs received title deeds by June 2023. TNUHDB made available title deeds for the remaining PAFs to sign but the legal heirs of the remaining five families have not yet done so because of deaths in these families. The World Bank continues to monitor this matter until it is resolved. The ULBs and TNUIFSL managed temporary adverse livelihood or economic impacts using the ESMF and RPF, and subproject specific ESIAs and ESMPs. The ESMPs were implemented to ensure timely action



and compensation for two severe incidents. All incidents were investigated in line with World Bank procedures and registered in the Fatality Tracking System of the World Bank. Families of the victims received prompt remedial actions and compensations. All incidents were closed except one that occurred in January 2023. The contractor offered compensation to the victim's family, but they sought judicial resolution to determine the compensation amount. This process is underway following procedures (ICR, paragraph 68).

Each ULB established a Grievance Redress Mechanism (GRM) as planned. PAFs accessed GRM through multiple channels including phones, websites, and ULB offices. Under the Chennai storm water drain subproject, petitions received were resolved by October 2016. Additional complaints were received in March 2018 regarding compensation for partially affected structures. These were subsequently settled. Compensation was paid. ULBs reported complaints related to the quality of construction and repairs of streets. These were made directly to contractors and resolved by them (ICR, paragraph 67). The government reported that they updated and will continue to use the ESMF to meet the safeguard requirements of other externally financed projects.

b. Fiduciary Compliance

Financial Management. The project complied with World Bank financial management policies (ICR, paragraph 70). ULBs kept up-to-date accounting records, submitted unaudited interim financial and audit reports with unqualified (clean) opinions on time before the COVID-19 pandemic. During the last two years of implementation, the ULBs experienced some delays in submitting their completed audit reports. In the first two years of implementation, administrative sanctions caused delays in releasing funds because the DMA lacked qualified staff to oversee financial management of the project. The project relied on the staff of another ongoing Bank-financed project, the Tamil Nadu Urban Infrastructure Financial Services Limited (TNUIFSL), to manage financial management of the project on a part-time basis.

Procurement. Most transactions complied with the World Bank procurement policies and procedures (ICR, paragraph 69). The project awarded most of the urban investment component contracts early in the project. TNUIFSL managed consultancies efficiently. The COVID-19 pandemic disrupted implementation and caused delays. Larger national/international construction firms did not participate in the investment packages, indicating barriers to entry. Procurement post reviews showed red flags. TNUIFSL and other government agencies lacked effective procurement supervision. The risk mitigation measure use of E-procurement was delayed. Since most of the TA component contracts were awarded outside of the World Bank's Systematic Tracking of Exchanges in Procurement (STEP) tool the World Bank team could not conduct post procurement review. No outstanding procurement issues were reported.

c. Unintended impacts (Positive or Negative)

d. Other



11. Ratings

Ratings	ICR	IEG	Reason for Disagreements/Comment
Outcome	Satisfactory	Satisfactory	
Bank Performance	Satisfactory	Satisfactory	.
Quality of M&E	Substantial	Substantial	
Quality of ICR	---	Substantial	

12. Lessons

The following five lessons emerged from the operation (ICR, paragraph 77-81):

- **Political commitment, policy reform, and provision of incentives are useful to improve the financial sustainability of water services.** In this project, water tariff in Tamil Nadu was low and not based on consumption. Data from the Supervisory Control and Data Acquisition System (SCADA) in Namakkal showed that water consumption level was similar between households covered by the 24/7 pilot and households receiving intermittent supply. ULBs need incentives to approve bylaws for volumetric billings to raise tariffs and increase collection. Metering could be piloted in selected areas where high levels of services and high-water consumption are expected. Proper systems could be in place to monitor consumers' usage and address complaints. Data can then be generated to inform ULBs' decision-making and counter public misconception on volumetric billing.
- **Information, education, and communication activities instill behavioral changes to sustain desired levels of services.** In this project, residents were encouraged to participate in O&M planning of municipal assets. However, there was no comprehensive communication strategy for the duration of the project. This could have been used to engage local communities, raise awareness, and enhance ownership of project investments. Consumers could be provided a reliable schedule of water supply and be educated on benefits of volumetric billing to encourage reducing wasting water. Overall, citizen participation could be further explored, including mechanisms for people to contribute in terms of time and labor.
- **The municipal development fund model may facilitate access to private market resources.** In this project, TNUDF provided loans and grants of over US\$3.2 billion in urban investments. ULBs registered 100 percent loan repayments. Future projects can support TNUDF to resume its pooled municipal bond program to increase ULBs' access to financial markets, with the funds raised giving priority to green projects such as sustainable water management, climate change adaptation, and energy efficiency.
- **Results-based performance grants help promote urban governance reform in ULBs.** In this project, the pilot Model Cities component demonstrated the effectiveness of linking fund allocation to support improved urban governance and citizen engagement, strengthening staffing and technical expertise, mobilization of own resources, and enhancing accountability in delivering services to city residents.



- **Institutional strengthening and capacity building of ULBs require sustained support.** ULBs have benefited from the various TA, training, and capacity support activities under the project. However, dedicated funding, staffing, and collaboration with training institutions are required to continue developing and delivering structured capacity building programs to strengthen institutional capacities of ULBs.

13. Assessment Recommended?

No

14. Comments on Quality of ICR

The ICR followed the guidelines and provided a detailed overview of the project. The theory of change helped to justify the reported ratings. Various parts of the report were mutually reinforcing and the ratings consistent. The report was results oriented, highlighting, for example, the measures adopted to ensure that the objectives remained achievable with the dropping of sub-components like CERF and the Hosur sewerage subproject. The report was candid, acknowledging shortcomings in risk assessments related to capacity of implementing agencies to undertake GIS related activities and the resulting unmet targets. Lessons were based on the project experience. The narrative and available evidence supported the ratings. The report is aligned to the project development objectives presenting data to support the ratings. The analysis of efficacy to achieve the first objective had minor shortcomings in the completeness of indicators that addressed the development problems of ULBs.

a. Quality of ICR Rating

Substantial